UNIT IV

17. Consider the following page reference string 1, 2, 3, 2, 5, 6, 3, 4, 6, 3, 7, 3, 1, 5, 3, 6, 3, 4, 2, 4, 3, 4, 5, 1. How many page faults would occur for the following replacement algorithm? Assume four frames and all frames are initially empty.

    (a) LRU replacement

    (b) FIFO replacement

    (c) Optimal replacement

Or

18. List out the file attributes and discuss the file access methods in detail.

UNIT V

19. Consider a disk queue with requests for I/O to blocks on cylinders 98,183,37,122,14,124,65,67. If the disk head starts at 53, then find out the total head movement with respect to FCFS, SSTF, SCAN and Look Scheduling.

Or

20. Compare and contrast the various issues in security related to Linux and windows XP operating system and explain how an application uses memory in windows XP.

B.Tech. DEGREE EXAMINATION, MARCH 2021.

Fifth Semester

COMPUTER SCIENCE AND ENGINEERING

OPERATING SYSTEMS

(2013 – 14 onwards)

Time : Three hours　　　　Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions

All question carry equal marks.

1. Mention the types of real time system.

2. List the services provided by operating system.

3. Under what circumstances user level thread is better than the kernel level thread?

4. How Synchronization can be achieved by monitor?

5. What are the advantages of dynamic loading?

6. What is the use of overlays?

7. How does the system detect thrashing?

8. Give any two criteria to choose a file organization.

9. What is meant by prefetching?

10. What is the responsibility of kernel in Linux operating System?

PART B — (5 × 11 = 55 marks)

Answer ALL question, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss the essential properties of Distributed system and the; types of System calls with an example for each.

Or

12. State the various criteria of a good process scheduling algorithm and discuss the major activities of an operating system in regard to process management.

UNIT II

13. Show how wait ( ) and signal ( ) semaphore operations could be implemented in multiprocessor environments, using the Test and Set ( ) instruction. The solution should exhibit minimal busy waiting. Develop Pseudocode for implementing the operations.

Or

14. Explain the FCFS, preemptive and non preemptive versions of Shortest - Job - First and Round Robin (time slice=2) scheduling algorithms with Gantt chart for the four processes given. Compare their average turn around and waiting time.

Process	Arrival Time	Burst Time
P1	0	10
P2	1	6
P3	2	12
P4	3	15

UNIT III

15. Compare paging with segmentation in terms of the amount of memory required by the address translation structures in order to convert virtual addresses to physical addresses.

Or

16. Is it possible to have deadlock involving only one process? Justify your answer and explain the necessary conditions to avoid deadlock.

5635130

B.Tech. DEGREE EXAMINATION, JANUARY 2022.

Fifth Semester

Computer Science and Engineering

OPERATING SYSTEMS

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Describe the operating system functions.

2. What are the objectives of operating system?

3. Define Semaphores.

4. What are the requirements that a solution to the critical section problem must satisfy?

5. What are the conditions under which a deadlock situation may arise?

6. Differentiate paging and segmentation.

7. Define demand paging in memory management.

8. How does the system discover thrashing?

9. What are the various disk-scheduling algorithms?

10. Define LINUX Virtualization.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. Explain different operating system structures with neat sketch.

Or

12. Discuss Inter Processor Communication in detail.

UNIT II

13. Suppose that the following processes arrive for execution at the times indicated. Each process will run the listed amount of time. In answering the questions, use non-preemptive scheduling and base all decisions on the information you have at the time the decision must be made.

Or

14.

Process	Arrival Time	Burst Time
P1	0.0	8
P2	0.4	4
P3	1.0	1

(a) Find the average turnaround time for these processes with the FCFS scheduling algorithm?

(b) Find the average turnaround time for these processes with the SJF scheduling algorithm.

UNIT III

15. Compare paging with segmentation in terms the amount of memory required by the addre translation structures in order to convert virtu addresses to physical addresses

Or

16. Explain different methods to handle and recov deadlocks.

UNIT IV

17. When page faults will occur? Describe the actio taken by operating system during page fault.

Or

18. Explain about file system mounting in detail.

UNIT V

19. What are the various disk space allocati methods. Explain in detail.

Or

20. Explain the following: (a) RAID (b) I/O in Linux.

5635130

B.Tech. DEGREE EXAMINATION, DECEMBER 2023.

Fifth Semester

Computer Science And Engineering

OPERATING SYSTEMS

(2013 – 14 Regulation)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

1. What is an Operating System?
2. Define Process.
3. What is a thread?
4. What is critical section?
5. What is swapping?
6. How to recover from deadlock?
7. What is the system call used to create a process?
8. What is a page?
9. List the major attributes and operations of a file system.
10. What is free space management?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

UNIT – I

11. (a) Describe system calls and system programs in detail. (7)

(b) What is a time-sharing system? Explain. (4)

Or

12. Explain in detail about Process Scheduling.

UNIT – II

13. Consider the following set with the length of CPU-burst time given in ms

Process	Burst Time	Arrival Time
P1	8	0
P2	4	1
P3	9	2
P4	5	3
P5	3	4

Draw four Gantt charts illustrating the execution of these -processes using FCFS, SJF, Priority and RR (quantum = 2 ms) scheduling. Also calculate waiting time and turnaround time for each scheduling algorithms.

Or

14. Outline a solution using semaphores to solve a dining philosopher problem.

UNIT – III

15. Explain in detail about Paging with segmentation.

Or

16. Explain deadlock avoidance algorithm with an example.

UNIT – IV

17. What is demand paging? Describe the process of demand paging in OS.

Or

18. What are the different types of directory structures? Explain them in detail.

UNIT – V

19. What is disk scheduling? Explain the different types of disk scheduling algorithms in detail.

Or

20. Discuss in detail about file system implementation.

UNIT V

19. Compare the functionalities of FCFS, SSTF, C-SCAN and CLOOK with example.

Or

20. Explain about Linux kernel and virtualization with neat sketch.

B.Tech. DEGREE EXAMINATION,
SEPTEMBER 2021.

Fifth Semester/ Third year

Computer Science and Engineering

OPERATING SYSTEMS

Time : Three hours | Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an operating system?

2. What are disadvantages of multi-processor systems?

3. What are the three different types of scheduling queues?

4. Define critical section.

5. Define Deadlock.

6. What is the purpose of paging the page table?

7. What is demand paging?

8. What is the basic approach for page replacement?

9. What are the allocation methods of a disk space?

10. What are the Components of a Linux System?

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the various types of system calls with examples.

Or

12. Discuss about

(a) Mainframe Systems

(b) Clustered Systems

(c) Multiprocessor Systems

UNIT II

13. Write notes about multiple-processor scheduling and real-time scheduling.

Or

14. State critical section problem? Discuss three solutions to solve the critical section problem.

UNIT III

15. Explain about given memory management techniques.

(a) Partitioned allocation

(b) Paging and translation look-aside buffer

Or

16. Explain in detail about deadlock avoidance algorithm.

UNIT IV

17. Consider the following page reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. How many page faults would occur for the following replacement algorithms, assuming three frames that all frames are initially empty?

Or

18. What are files and explain the access methods for files?